TASK 1 REPORT

DNA/Protein Sequence Analysis of Human BACE1 Protein Associated with Alzheimer's Disease

Abstract

Bioinformatics tools provide powerful approaches for studying protein evolution and function. In this study, the human Beta-site APP Cleaving Enzyme 1 (BACE1) protein, a key therapeutic target in Alzheimer's disease, was analyzed using the BLASTp algorithm. The protein sequence was retrieved from NCBI and compared against protein databases to identify homologous sequences. Results revealed highly conserved BACE1 proteins among primates and mammals, indicating strong evolutionary conservation and biological importance.

Introduction

Alzheimer's disease is a progressive neurodegenerative disorder characterized by memory loss and cognitive decline. One of the major pathological features of Alzheimer's disease is the accumulation of amyloid-beta (Aβ) plaques.

BACE1 (Beta-site APP Cleaving Enzyme 1) initiates the cleavage of Amyloid Precursor Protein (APP), leading to Aβ peptide formation. Due to its role in amyloidogenesis, BACE1 has become an important target for drug discovery and therapeutic intervention.

Sequence similarity analysis provides insights into protein conservation, evolutionary relationships, and functional importance.

Objective

To identify homologous proteins related to human BACE1 using BLASTp and evaluate sequence conservation across species.

Methodology

- **Sequence Retrieval:** The amino acid sequence of human BACE1 was retrieved from the NCBI protein database in FASTA format.
- **BLAST Analysis :** Protein BLAST (BLASTp) was performed against the non-redundant protein sequence database.

Parameters:

- Program: BLASTp
- Database: Non-redundant protein sequences (nr)
- Query Sequence: Human BACE1
- Organism Filter: None

The resulting homologous sequences were analyzed based on:

- Query Coverage
- Percentage Identity
- Alignment Score
- E-value

Results

Top Homologous Proteins Identified

Species	Protein
Homo sapiens	BACE1
Cebus imitator	BACE1
Pongo abelii	BACE1
Callithrix jacchus	BACE1
Gorilla gorilla gorilla	BACE1
Macaca mulatta	BACE1

Conserved Domain Analysis

The BLAST graphical summary revealed the presence of:

- Aspartyl protease catalytic domain
- Active site flap region
- Pepsin-retropepsin-like superfamily domain

These domains are characteristic of BACE1 enzymes and are essential for proteolytic activity.

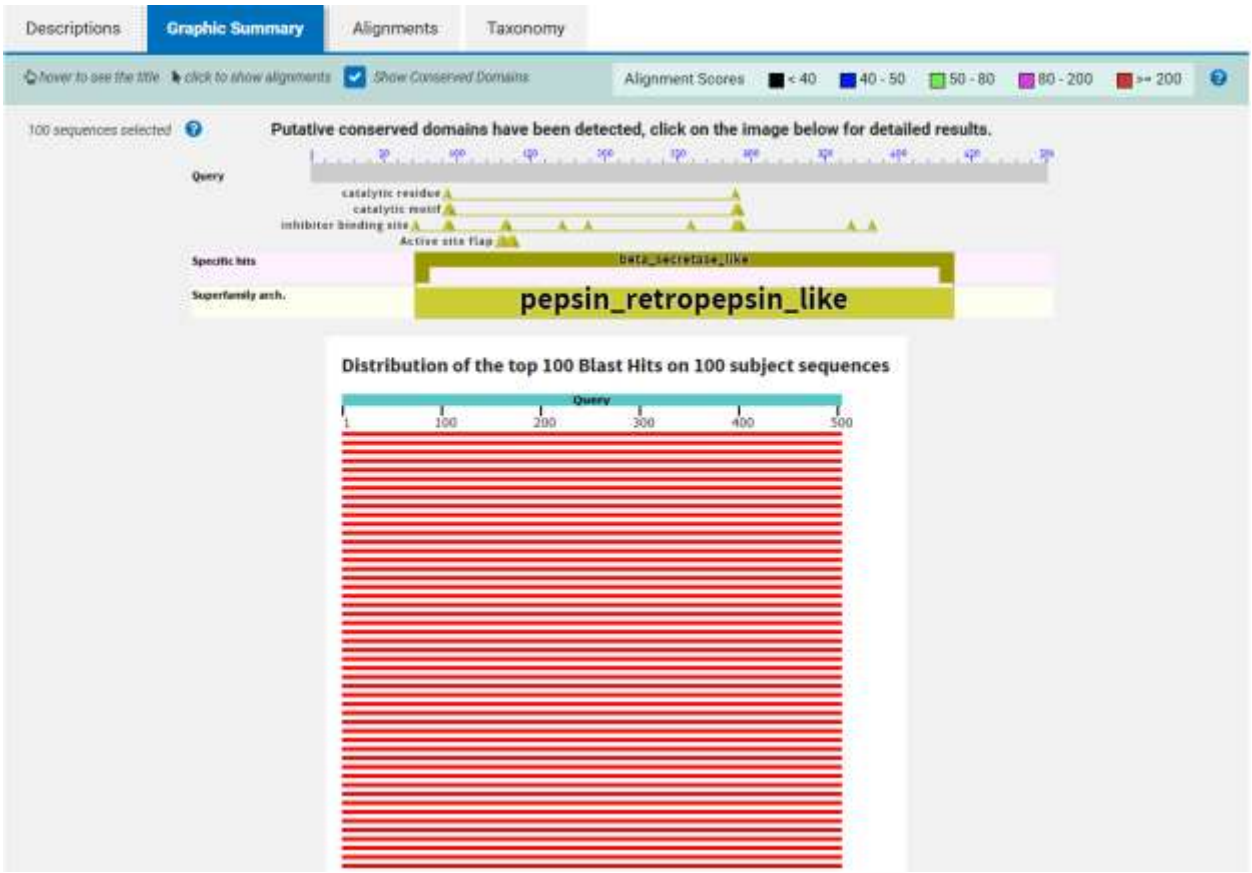


Figure 1

BLAST graphical summary showing highly conserved BACE1 homologs and conserved pepsin-retropepsin-like domains among mammalian species.



Figure 2

Representative homologous BACE1 protein sequences identified through BLAST analysis.

Discussion

BLAST analysis demonstrated extensive conservation of BACE1 among primate species. The high sequence similarity suggests preservation of enzymatic function throughout evolution.

The conserved catalytic aspartyl protease domains observed in all homologs indicate that BACE1 performs similar biological roles across species. Such conservation highlights the importance of BACE1 in cellular processes and validates its use as a target for Alzheimer's disease therapeutics.

The presence of homologous sequences in both humans and non-human primates also supports the use of animal models for studying Alzheimer's disease mechanisms and evaluating novel inhibitors.

Conclusion

The BACE1 protein exhibits strong evolutionary conservation among mammalian species. BLAST analysis identified multiple homologous proteins possessing highly conserved catalytic domains. These findings reinforce the biological importance of BACE1 and its relevance as a therapeutic target in Alzheimer's disease research.

References

- Altschul SF et al. (1997). Gapped BLAST and PSI-BLAST. Nucleic Acids Research.
- National Center for Biotechnology Information (NCBI).
- UniProt Consortium (2025).
- Vassar R. (2022). BACE1 in Alzheimer's Disease.